EXPERIMENT-10

Name: Sruthi Dinesh
Roll No. : 240701536
Dept. : Computer Science and Engineering
Subject: User Interface Design
Subject Code: CS23A34

AIM:
The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:
Step 1: Set Up Your HTML File
First, create an HTML file to hold your canvas for the chart and include Chart.js

```html

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Inventory Data Visualization</title>
  <link rel="stylesheet" href="style.css" />
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
</head>
<body>
  <div class="container">
    <h1>Inventory Management Dashboard</h1>

    <div class="chart-box">
      <h2>Inventory Category Distribution</h2>
      <canvas id="pieChart"></canvas>
    </div>

    <div class="chart-box">
      <h2>Stock Levels by Item</h2>
      <canvas id="barChart"></canvas>
    </div>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

Step 2: Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic.

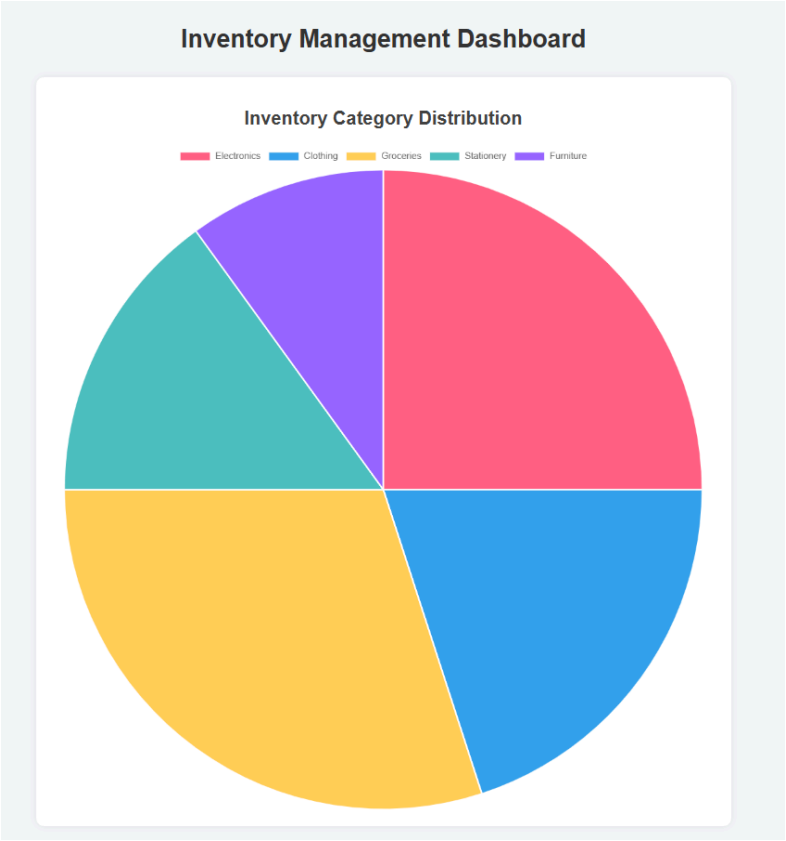
```javascript
const categoryData = {
  labels: ["Electronics", "Clothing", "Groceries", "Stationery", "Furniture"],
  datasets: [
    {
      data: [25, 20, 30, 15, 10],
      backgroundColor: ["#ff6384", "#36a2eb", "#ffcd56", "#4bc0c0", "#9966ff"]
    }
  ]
};

const stockData = {
  labels: ["Laptop", "T-Shirt", "Rice Bag", "Notebook", "Chair"],
  datasets: [
    {
      label: "Stock Quantity",
      data: [50, 120, 80, 150, 40],
      backgroundColor: "#36a2eb"
    }
  ]
};

const pieCtx = document.getElementById("pieChart").getContext("2d");
new Chart(pieCtx, {
  type: "pie",
  data: categoryData,
  options: {
    responsive: true
  }
});

const barCtx = document.getElementById("barChart").getContext("2d");
new Chart(barCtx, {
  type: "bar",
  data: stockData,
  options: {
```

```javascript
});

const barCtx = document.getElementById("barChart").getContext("2d");
new Chart(barCtx, {
  type: "bar",
  data: stockData,
  options: {
    responsive: true,
    scales: {
      y: {
        beginAtZero: true
      }
    }
  }
});
```

Step 3:
Set Up Your CSS File

```
body {
  font-family: Arial, sans-serif;
  background-color: #f4f6f9;
  margin: 0;
  padding: 20px;
}

.container {
  max-width: 900px;
  margin: auto;
  text-align: center;
}

h1 {
  margin-bottom: 30px;
  color: #333;
}

.chart-box {
  background: white;
  margin-bottom: 30px;
  padding: 20px;
  border-radius: 10px;
  box-shadow: 0 0 10px rgba(0, 0, 0, 0.08);
}

h2 {
  margin-bottom: 20px;
  color: #444;
}

canvas {
  max-width: 100%;
}
```

OUTPUT :


Inventory Management Dashboard
Inventory Category Distribution
Electronics
Clothing
Groceries
Stationery
Furniture

Stock Levels by Item
Stock Quantity
160
140
120
100
80
60
40
20
0
Laptop
T-Shirt
Rice Bag
Notebook
Chair